METRIC

A-A-52565
5 March 1997

# COMMERCIAL ITEM DESCRIPTION
# SURVEYING INSTRUMENT; AZIMUTH GYRO, LIGHTWEIGHT (SIAGL)

The General Services Administration has authorized the use of this Commercial Item Description (CID) as a replacement for MIL-S-52256 for all federal agencies.

1. SCOPE This CID covers an automated north seeking gyroscopic survey system.

2. SALIENT CHARACTERISTICS

2.1. Description. The survey system, hereinafter referred to as "SIAGL" shall consist essentially of an automatic azimuth north seeking gyroscope, integrated electronics and control panel, battery power pack, top mounted scale reading theodolite and a telescopic leg tripod with provisions for mounting the assembled instrument.

2.2 Physical characteristics.

2.2.1 Azimuth gyroscope. The gyroscope shall be capable of automatically determining azimuths relative to geographic north or grid north for display readout or remote integration through an RS232 interface or use industry standard data storage media and high speed data transfer. The unit shall consist of a gyroscope motor, electronic operator control and display panel, battery pack or power unit, and internal electrical power circuitry for both gyroscope and theodolite illumination. The gyroscope shall have means for accepting a top mounted theodolite and a battery pack or power unit. The gyroscope shall be adaptable to a telescopic leg tripod. A means for plumbing shall be provided on the base of the gyroscope housing. The gyro assembly shall not be damaged if overturned.

2.2.2 Theodolite. The theodolite shall be an optical reading directional instrument or an auto-reading/recording digital instrument. The theodolite shall measure/display horizontal and vertical

Beneficial comments, recommendations, additions, deletions, clarifications etc., and any other data which may improve this document should be sent to: U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN CETEC-PD-TE, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864.

AMSC N/A FSC 6675

Distribution statement A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

angles at the precision and repeatability required for third order accuracy. The instrument shall be graduated in mils. It shall weigh less than 5.9 kg.

2.2.2.1 Telescope. The telescope shall produce an erect image. The telescope shall have an aiming device and sunshade. The stadia ratio shall be 1/100. The reticle shall consist of a pattern of opaque lines similar to figure 1.

2.2.2.2 Eyepiece. The eyepiece shall be equipped with a diopter scale marked with plus and minus units from zero, and shall have an index mark for referencing the setting. When specified (see 6.2), the following options shall be provided with the eyepiece:

a. An eyepiece solar filter with a density range between 5.0 and 6.0 that is attachable without tools.
b. An eyepiece prism which permits observation of ground points and angles within 26 degrees from the zenith with the telescope both in the direct and reversed positions.
c. An eyepiece prism filter with a density range between 5.0 and 6.0.
d. Telescope auto-collimation eyepiece that is attachable without tools. The magnification of the telescope with this eyepiece shall be not less than 18 diameters.

2.2.2.3 Circles. The circles shall read directly to 0.20 mil. The horizontal circle shall be mounted so that readings increase as the alidade is rotated clockwise. The root mean square (RMS) error of the circle shall not exceed 0.20 mil at ambient temperature of approximately 21° C. The horizontal circle shall have a means of interpolating readings smaller than the circle graduations. The vertical circle shall be graduated and numbered so that zero represents zenith and a reading of 1,600 mil is obtained when the telescope is level and in the direct position. A means shall be provided to view the horizontal and vertical circle measurements, simultaneously. The vertical and horizontal circle measurements shall be identified.

2.2.2.4 Levels. The theodolite shall be equipped with a leveling capability which allows the instrument to be set and operated at the required accuracies.

2.2.2.5 Altitude self-indexing device. The altitude self-indexing device shall be damped so that oscillation is stopped in not greater than 5 seconds. The device shall be accurate to within 0.06 mil or 12 arc seconds. The accuracy of the pendulum is 1 arc second over a range of ±6 arc minutes.

2.2.2.6 Target rod. When specified, a target rod shall be furnished. The rod shall be adaptable to the top of the theodolite. The target rod shall be cylindrical in shape with a provision to light the top of the rod for nighttime operations. The illumination color shall be changeable. The top of the rod shall be within ±1.5 mm of the projected vertical axis of the theodolite. Power for the illumination of the target rod shall be supplied by the SIAGL system power source.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

2.2.3 Microprocessor and control/display panel.

2.2.3.1 Microprocessor. The microprocessor shall have a nonvolatile memory with sufficient capacity to store all data, constants, operating parameters, and sufficient random access memory to perform required data reductions. Calibration data sheets shall be furnished with each SIAGL.

2.2.3.2 Control/display panel. The panel shall contain all function keys, an alpha-numeric display unit, a receptacle to accommodate both RS232 and interface with the Global Positioning System (GPS). All keys shall be clearly identified as to function. Receptacles shall be labeled according to function. Warning and indicator lights required for operation of the SIAGL shall also be located on the panel.

2.2.4 Power and illumination

2.2.4.1 Power Source. The SIAGL shall not require more than 35 watts, peak power, when operating from any of the following sources:

a. Battery power pack
b. 115/230 volt AC 50/60 Hz utility power source for battery charging
c. Vehicle power through DC power adapter

2.2.4.2 Battery power pack. Two battery packs shall be furnished. Each battery pack shall have sufficient ampere-hour capacity for the SIAGL to accurately conduct not less than 40 azimuth determinations at a 21° C ambient temperature. The pack shall be mounted on and adapted to the azimuth gyroscope in such a manner as to require no external wiring. The battery pack shall be easily removable for storage or exchange. The battery power pack shall be designed to prevent reverse polarity installation.

2.2.4.3 Auxiliary Power. The SIAGL shall operate from external 12 volt DC batteries providing the same operating capabilities as with the internal power source. An external power cable shall be provided with the system that shall not be less than 6 m in length and shall terminate in automotive type battery clamps.

2.2.4.4 Battery charger. System batteries shall be rechargeable using a charger supplied with the system. The battery charger shall be operable from 115 volt, 60 Hz, or 220 volt, 50 Hz AC power. The battery charger shall be capable of concurrently charging at least two battery packs or a built-in battery charger shall be furnished with each battery pack. The battery shall not be damaged or loose capacity when charged continuously from the charger supplied with the system.

2.2.4.5 Cables and connectors. External power cables shall be provided for operating the SIAGL and charging the batteries. One each of the following power cables shall be furnished:

a. Cable from a vehicle power source for charging SIAGL batteries, not less than 4.57 m long.
b. Cable from a 115 volt, 60 Hz, or 220 volt, 50 Hz AC power source to battery charger, not less than 1.83 m long.
c. Cable from a vehicle power source to power a conditioner or adapter when used in lieu of regular battery power pack, not less than 7.62 m long.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

2.2.4.6 Illumination. The SIAGL shall be illuminated by both natural and artificial means. The artificial illumination shall be adjustable. When the SIAGL is operated in the dark there shall be no light visible beyond 33 meters except that within the telescope field of view.
Power for illumination shall be integral or provided through a cable.

2.2.5 Built-in test. The SIAGL shall be provided with a built-in self-test for use prior to making azimuth determinations. The built-in self-test system shall be capable of detecting and locating failures.

2.2.6 Tripod. A collapsible type, telescopic leg, tripod shall be furnished. The tripod shall conform to A-A-52298 except that it shall be provided with a 3-point, standard screw type, adjustable plate assembly. The assembly shall have a circular center opening of sufficient size to permit a 2.54 cm minimum diametrical adjustment for the SIAGL, over an established ground point, and a circular level for rough leveling and SIAGL set-up. The adapter plate, when mounted on the tripod, shall be included in the overall height of the tripod. The tripod shall have an adjustable height range from 89 ±5 to 135 ±5 cm.

2.2.7 Instrument cover. A protective cover for the SIAGL shall be provided.

2.2.8 Winterization kit. The winterization kit shall include covers for all control knobs and leveling screws (unless plastic or plastic coated), as well as shields for the telescope, microscope, and optical plummet eyepiece.

2.2.9 Transportability. The total weight of the SIAGL including battery power source, battery charger, and tripod with each major component and applicable accessories in their respective transport cases shall not exceed 63.56 kg. The weight of any individual transport case shall not exceed 20 kg when loaded. Each transport case shall be constructed of material(s) capable of protecting each component from shock, vibration and impact that may be encountered during transport. Adjustable carrying straps with attached cushion material shall be provided to permit backpacking each case individually or provision shall be made for the cases to be mounted on the Army packboard. Each case shall be permanently marked to indicate that it contains a delicate instrument and to show the proper orientation of the case for transport. A case is not required for the tripod.

2.3 Performance characteristics. The SIAGL shall determine azimuths at any operational latitude up to 70 degrees with an error in true azimuth determination not to exceed a 0.20 mil probability of error (PE) and an error not to exceed a 0.30 mil PE from 70 to 75 degree latitudes. The time required from power-on to determination of azimuths, of required accuracy, shall not exceed 5.0 minutes for all latitudes up to 75 degrees at ambient temperatures specified in 2.4.1. Pre-orientation or instrument warm-up shall not be required. The SIAGL shall permit removal from the operating site within 2 minutes without affecting the accuracy of the gyroscope sensor. The SIAGL shall be capable of not less than 100 missions of operation without recalibration, alignment or adjustment to a predetermined azimuth. The SIAGL shall be provided with a level vial of the required accuracy and sensitivity to permit the SIAGL to be set up and operated automatically without interruption.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

2.4 Environmental requirements. The SIAGL shall not be damaged nor performance degraded by the following conditions or combination, there of:

2.4.1 Operating temperature range. An ambient temperature range of 52° +3°,-0° C to -40° +0, -3° C.

2.4.2 Storage and transit temperature conditions. Storage or transit at temperatures as high as 68° C for 4 hours after stabilization and as low as -40° C for 3 days without solar radiation.

2.4.3 Wind. Winds of 9 ±1 m/s, with wind gusts to 11 ±1 m/s.

2.4.4 Rainfall.

2.4.4.1 Assembled Instrument. Wind driven rainfall of 5 ±3 cm per hour, from any direction, and at any angle from vertical to 45 degrees from vertical, for a period of 5 minutes ± 30 seconds.

2.4.4.2 Transport cases. Rainfall requirements of 2.4.4.1. In addition, rain and wind at velocities of 18 ±2 m/s to all sides of the case equally for a period of not less than one hour.

2.4.5 Blowing dust. A blowing dust environment while in the transport case with the following criteria:

a. The air velocities of 1.5 m/s to 8.9 m/s.
b. The dust composition of red china clay or silicon flour.
c. The dust concentration of 10.6 ±7 g/$m^3$.
d. The SIAGL orientation of all surfaces to the blowing dust.
e. The air temperature at 23° C for 6 hours and 50° C for 6 hours after stabilization.

2.4.6 Blowing sand. The SIAGL shall not be damaged by a blowing sand environment while in the transport case with the following criteria:

a. The air velocities of 18 to 29 m/s.
b. The sand composition (at least 95% silicate oxide by weight) and grain sizes varying from 20 mesh screen to 100 mesh screen.
c. The sand concentration of.2 ±0.50 g/$m^3$.
d. The SIAGL orientation of all surfaces to the blowing sand for 90 minutes per surface.
e. The air temperature of 50° C.

2.4.7 Vibration. Vibrations of 2.54 - 3.00 cm, double amplitude, orbital path at 5 Hz for 45 minutes on 3 axes while stored in the transit case.

2.4.8 Shock. A drop of 33 cm on a hard surface on each of the 6 faces, each of the 8 corners, and each of the 12 edges, for a total of 26 drops while stored in the transit case.

2.4.9 Magnetic field. In the presence of a magnetic field of 3 ± ½ gauss.

2.4.10 Fungus and moisture resistance. Exposure to fungi and moisture.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

2.5 Electromagnetic interference. The SIAGL shall have design characteristics for the control of electromagnetic interference emission and susceptibility.

2.5.1 Conducted emissions. Conducted emissions (CE) on power leads and returns, that obtain power from sources not part of the unit, shall not exceed the values in figure 2 for a frequency scan from 10 kHz to 10 MHZ.

2.5.2 Conducted susceptibility. Conducted susceptibility (CS) on all interconnecting cables, including power cables, shall not exceed the values in figure 3 for a frequency spectrum scan from 10 kHz to 10 MHZ. Use curve A for frequency ranges 10 kHz to 2 MHZ, and curve B for frequency ranges from 2 MHZ to 10 MHZ.

2.5.3 Radiated emissions. Radiated emissions (RE) from the SIAGL and all interconnecting cables shall not exceed the values in figure 4 for a frequency spectrum scan from 2 MHZ to 18 GHz. Testing is required up to 1 GHz or 10 times the highest intentionally generated frequency within the SIAGL, whichever is greater. Measurements beyond 18 GHz are not required.

2.5.4 Radiated susceptibility. Radiated susceptibility (RS) shall not exceed 50 volts/meter for a frequency spectrum scan from 2 MHZ to 18 GHz. The requirements shall be met for vertically polarized fields up to 30 MHZ and for both horizontally and vertically polarized fields above 30 MHZ. Circular polarized fields are not acceptable.

2.6. Maintenance ratio. The ratio of maintenance man-hours to the operating hours shall not exceed 0.05.

3. REGULATORY REQUIREMENTS The contractor/offeror is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS The supplier shall comply with the following for certification and acceptability.

4.1. Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

4.2. Market acceptability. The SIAGL offered shall be of the latest model of the standard product of the supplier and that has been used in the field for at least one year by commercial organizations.

4.3. Examination. Each SIAGL shall be examined for compliance with the requirements specified in 2.2 through 2.6. Any modification necessary following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects lessening maximum efficiency shall constitute cause for rejection.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

5. PACKAGING Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES

6.1. When this CID is used for procurement the product conformance clause should appear in the solicitation.

6.2 Acquisition Requirements.

a. Title, number, and date of this document.
b. When telescope eyepiece filter is required (see 2.2.2.2).
c. When telescope eyepiece prism is required (see 2.2.2.2).
d. When telescope eyepiece prism filter is required (see 2.2.2.2).
e. When telescope auto-collimation eyepiece is required (see 2.2.2.2).
f. When target rod is required (see 2.2.6).
g. Level of preservation and packing required.

6.3 Source of Documents. A-A-52298 is available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Ave, Philadelphia, PA 19120-5099

6.4 Definitions. The following definitions apply to this document:

6.4.1 Mil. Mil is a military unit of measure equal to the angle subtended by 1/6400 of a circumference.

6.4.2 Recovered materials. For the purpose of this requirement, recovered materials are those which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. Used, rebuilt, or remanufactured component, pieces and parts shall not be incorporated in the SIAGL.

6.4.3 Stabilization. Stabilization is a 2 degree Centigrade or less differential between the ambient temperature and the instrument.

6.5. Subject term (key word) listing.

theodolite
gyroscope



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

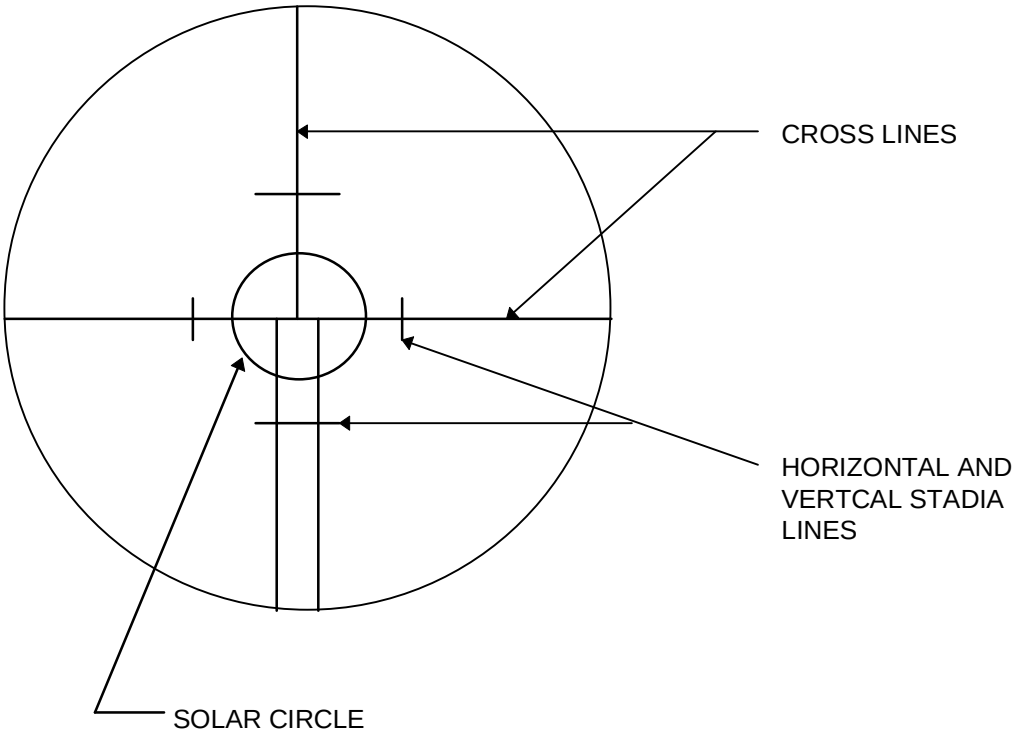



Figure 1. Reticle Pattern for Theodolite.
(Illustrative example only)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

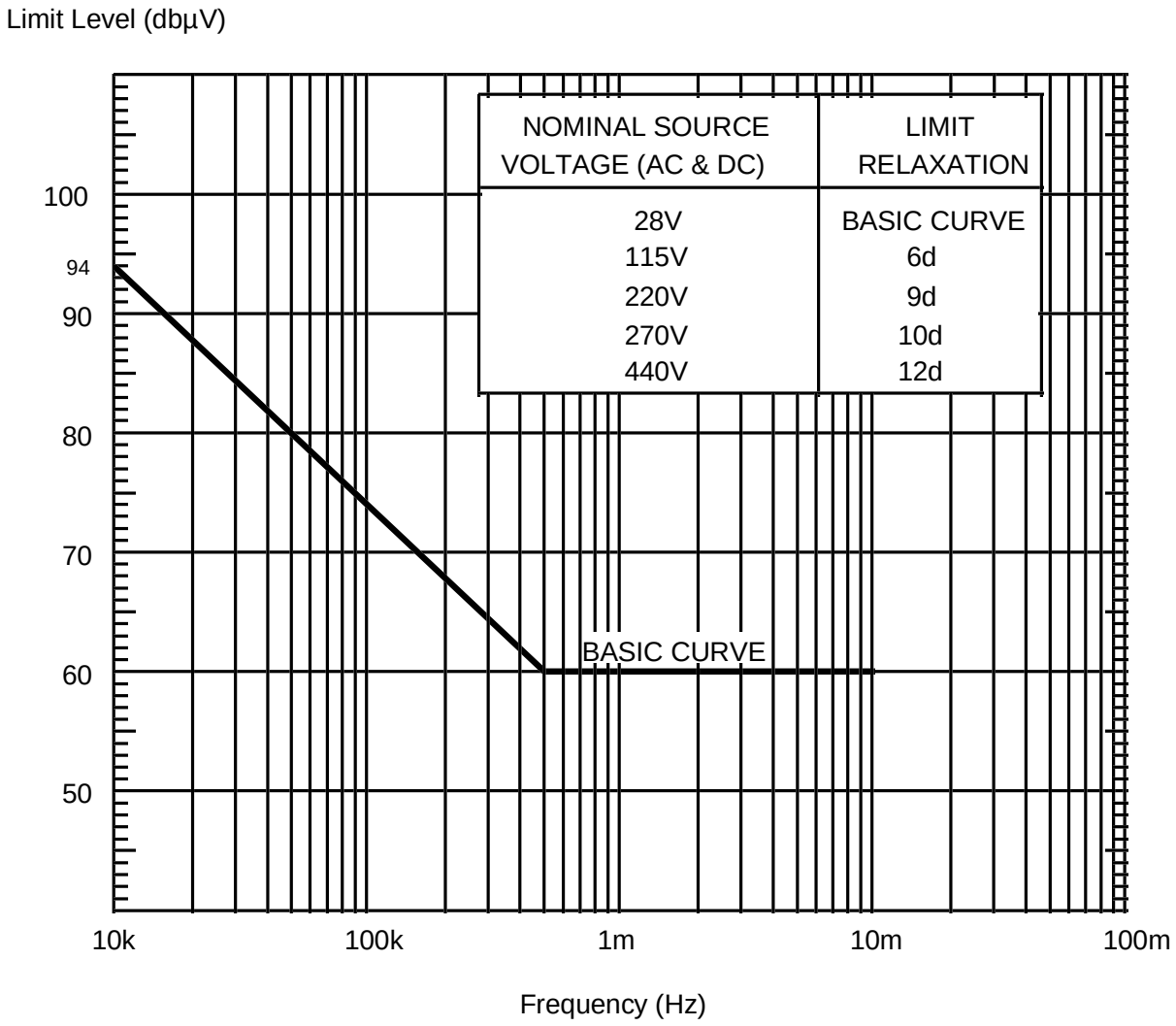



Figure 2. Conducted Emission Limit



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

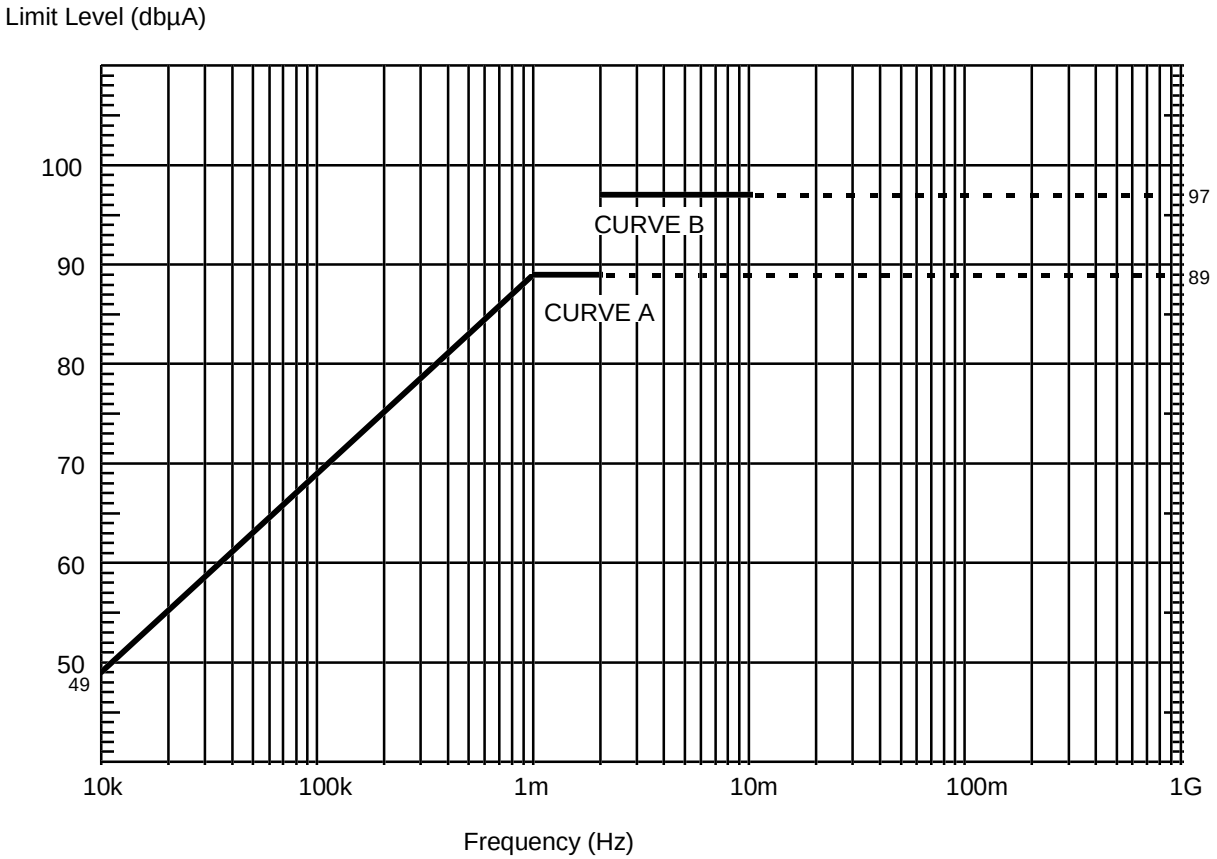



Figure 3. Conducted Susceptibility Limit



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

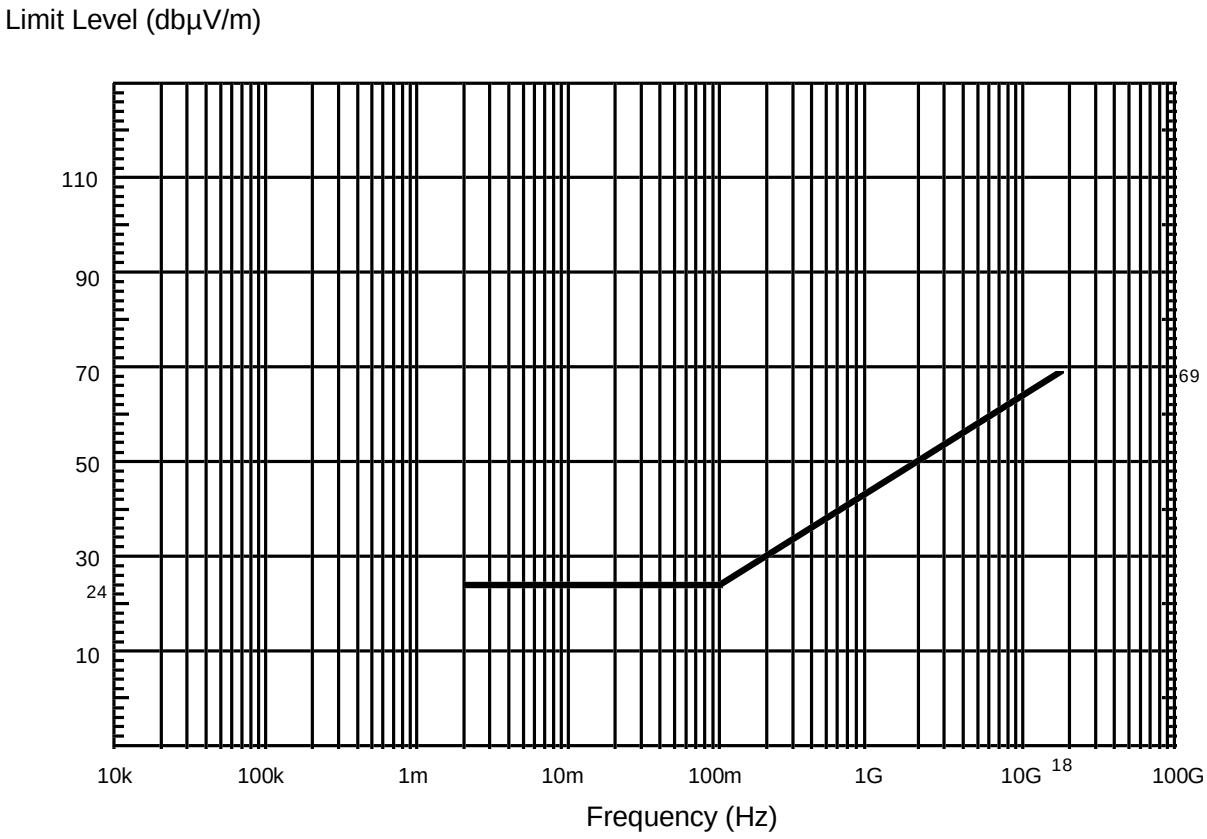



Figure 4. Radiated Emissions Limit



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CE

Review Activity:
DLA - GS

Civil Coordinating Activity:
GSA - FSS

Preparing Activity:
Army - CE
(Project 6675-0050)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:13Z
Check the source to verify that this is the current version before use.